Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

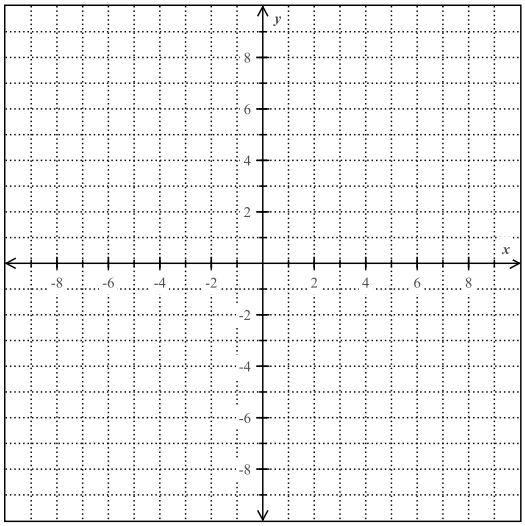
Number of Questions: **39**

Instructor Name: **Bodanske, Daniel**

Question 1 of 39

Graph the line.

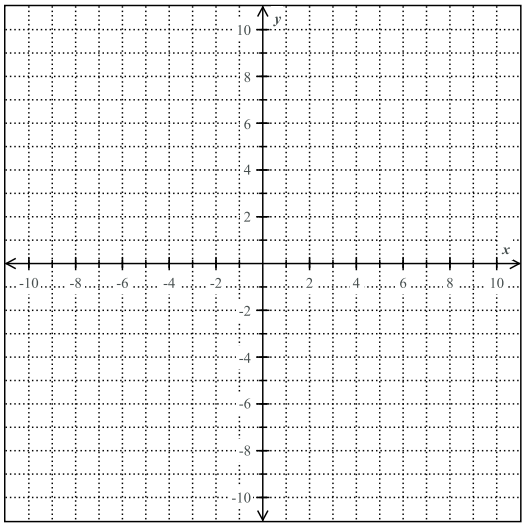
$$y = 3x - 7$$



Question 2 of 39

Graph the line.

$$y = -\frac{3}{5}x + 1$$



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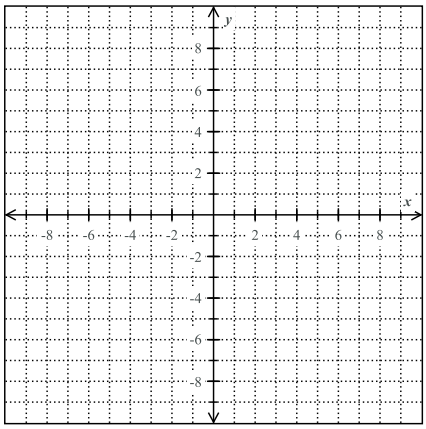
Question 3 of 39

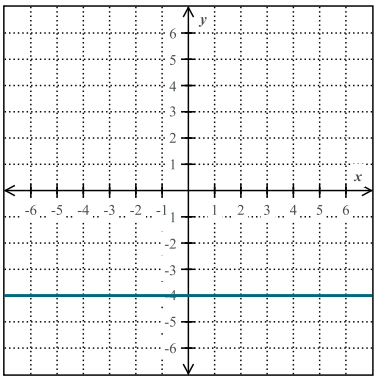
Graph the line.

$$x + y = 3$$



Question 4 of 39

Find the y-intercept and the x-intercept of the line below.



Question 5 of 39

Find the x-intercept and y-intercept of the line.

$$5x + 2y = -10$$

x-intercept: _____

y-intercept: _____

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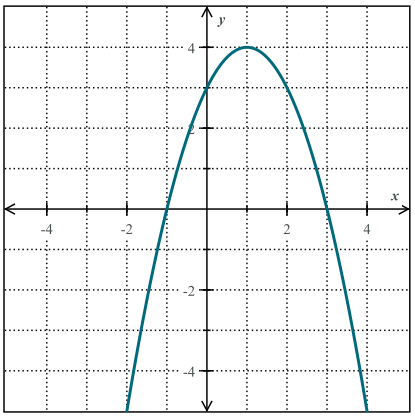
Question 6 of 39

The graph of a function is given below.
Give all x-intercepts and y-intercepts shown.



If there is more than one answer, separate them with commas.

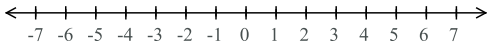
Click on "None" if applicable.

x-intercept(s): _____

y-intercept(s): _____

Question 7 of 39

Graph the set $\{x \mid -5 \leq x \leq 3\}$ on the number line.
Then, write the set using interval notation.



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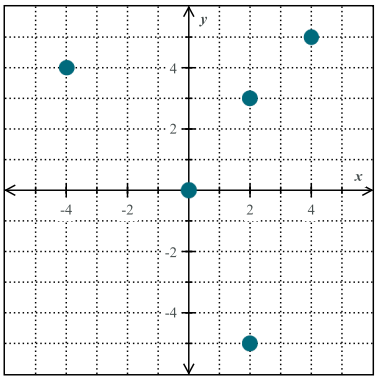
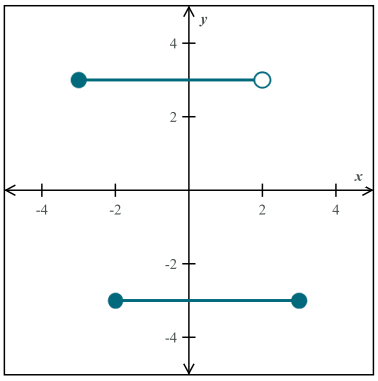
Question 8 of 39

Six graphs are shown below. Click on the tabs to see them.

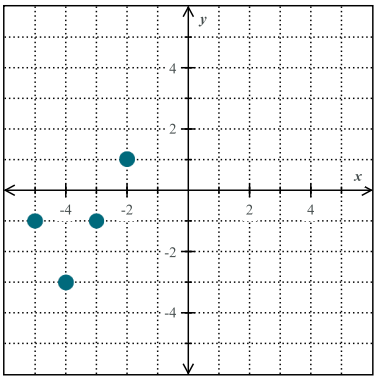
Graph 1



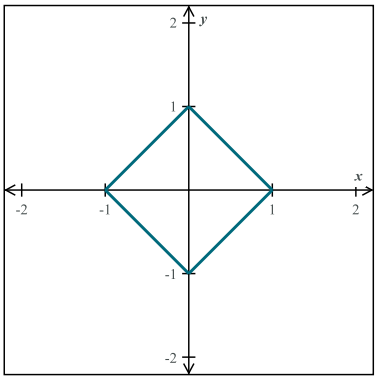
Graph 2



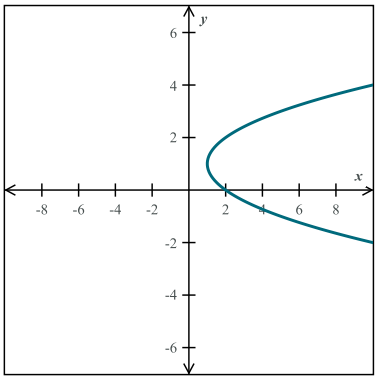
Graph 3



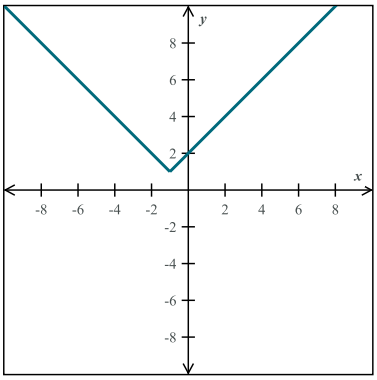
Graph 4



Graph 5



Graph 6



Check the box corresponding to each graph that represents a function.

☐ Graph 1 ☐ Graph 2 ☐ Graph 3 ☐ Graph 4 ☐ Graph 5 ☐ Graph 6 ☐ None of the graphs

Question 9 of 39

Suppose that the relation T is defined as follows.

$$T=\{(7,\ -2),\ (9,\ 0),\ (-9,\ -2)\}$$

ve the domain and range of T.
ite your answers using set notation.

domain: _____

range: _____

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Question 10 of 39

The function f is defined by the following rule.

$$f(x) = -4x - 5$$

Complete the function table.

x	$f(x)$
-3	
0	
3	
4	
5	

Question 11 of 39

The functions f and g are defined as follows.

$$f(x) = 2x - 1 \qquad g(x) = 2x^2 - 3x$$

Find $f(-5)$ and $g(-2)$.

Simplify your answers as much as possible.

Question 12 of 39

The function h is defined by $h(x) = x^2 + 7$.

Find $h(5x)$.

$h(5x) =$ _____

Question 13 of 39

A construction crew is lengthening a road that originally measured 57 miles. The crew is adding one mile to the road each day. The length, L (in miles), after d days of construction is given by the following function.

$$L(d) = 57 + d$$

What is the length of the road after 22 days?

_____ miles

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Question 14 of 39

A printing service charges a set-up fee of $\$15.25$ for each order and 9 cents more for each copy. The total cost, C (in dollars), for an order of x copies is given by the following function.

$$C(x) = 0.09x + 15.25$$

What is the total cost for an order of 50 copies?

_____ dollars

Question 15 of 39

Goran wants to save $\$700$ to buy a TV. He saves $\$19$ each week. The amount, A (in dollars), that he still needs after w weeks is given by the following function.

$$A(w) = 700 - 19w$$

Answer the following questions.

(a) How much money does Goran still need after 7 weeks?

$\$$.

(b) If Goran still needs $\$339$, how many weeks has he been saving?

_____ weeks

Question 16 of 39

A construction crew is lengthening a road. Let L be the total length of the road (in miles). Let D be the number of days the crew has worked. Suppose that $L = 3D + 300$ gives L as a function of D. The crew can work for at most 70 days.

Identify the correct description of the values in both the domain and range of the function. Then, for each, choose the most appropriate set of values.

	Description of Values	**Set of Values**
Domain:	○ number of days the crew has worked ○ length of the road (in miles)	Select
Range:	○ number of days the crew has worked ○ length of the road (in miles)	Select

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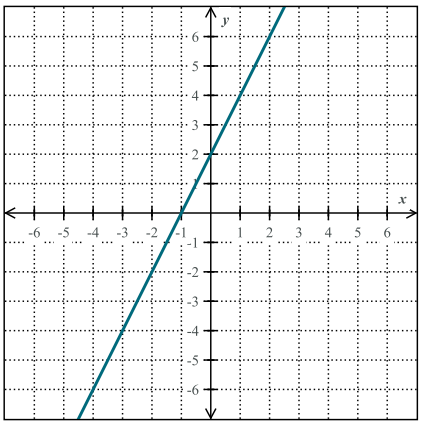
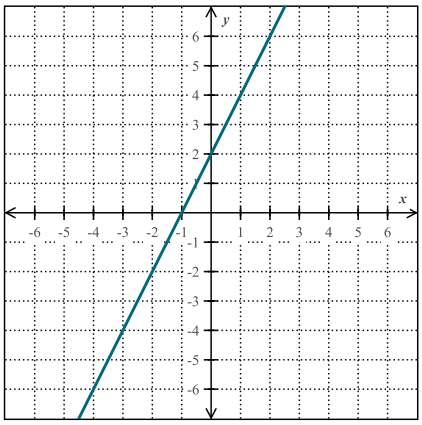
Question 17 of 39

The graph of a function h is shown below.
Find $h(-1)$.

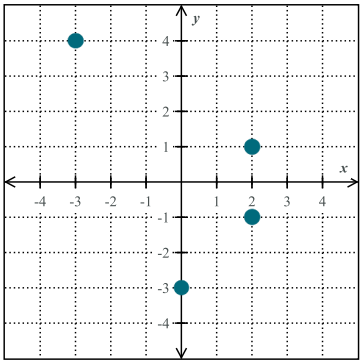




$h(-1) = \square$

Question 18 of 39

The graph of the relation G is shown below.



Give the domain and range of G.
Write your answers using set notation.

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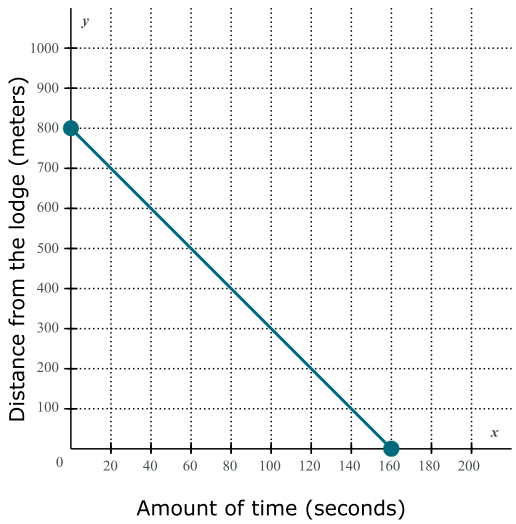
Question 19 of 39

Sam is riding a cable car to a lodge. His distance from the lodge is recorded from the start of the ride to the end.

Let y be the distance (in meters) from the lodge and let x be the amount of time (in seconds) since the ride started. The equation $y = -5x + 800$ models y as a function of x. As shown in the graph below, this model is accurate only from the start of the ride until the end. (The endpoints of the graph show when the ride starts and ends.)



Use the graph to complete each part.

(a) Sam is _____ meter(s) from the lodge at the start of the ride. He is _____ meter(s) from the lodge at the end of the ride.

Each value in the {domain, range} is a distance from the lodge.

(b) Sam starts the ride when the amount of time is _____ second(s). He continues riding until the amount of time is _____ second(s).

Each value in the {domain, range} is an amount of time since the ride started.

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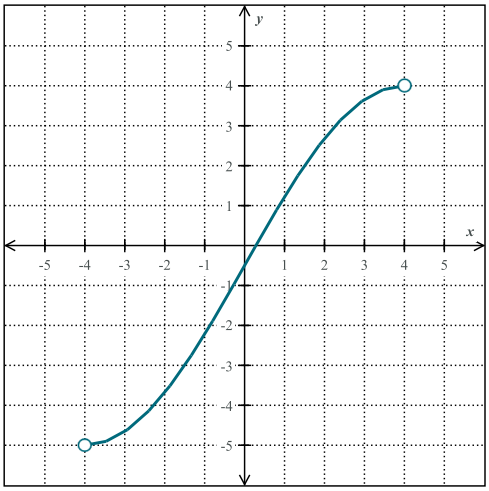
Question 20 of 39

The entire graph of the function f is shown in the figure below.



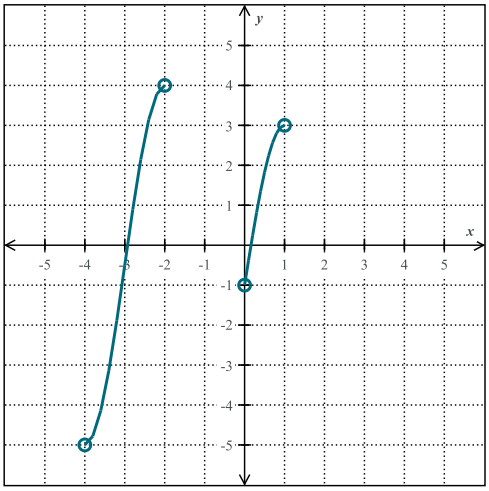
Write the domain and range of f using interval notation.

domain: _____

range: _____

Question 21 of 39

The entire graph of the function g is shown in the figure below.
Write the domain and range of g as intervals or unions of intervals.



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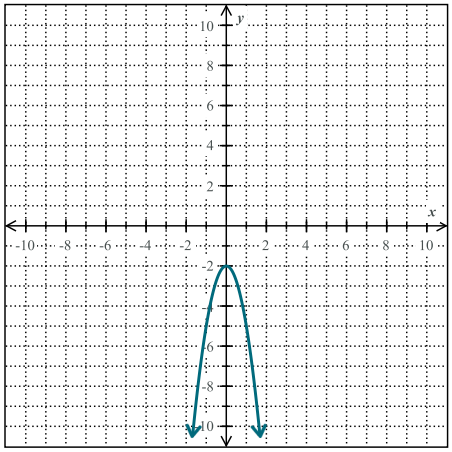
Question 22 of 39

The graph of a quadratic function with vertex $(0,\ -2)$ is shown in the figure below.
Find the range and the domain.

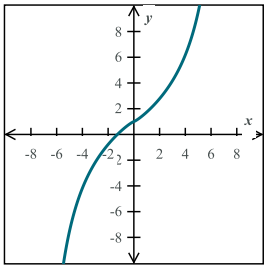
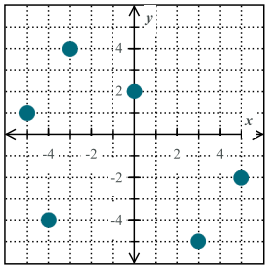
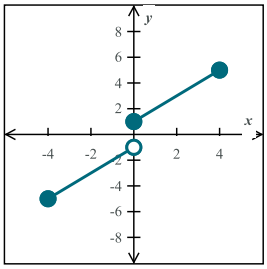
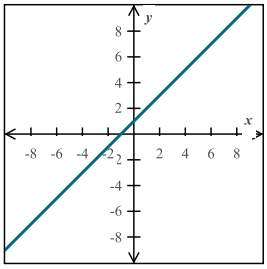
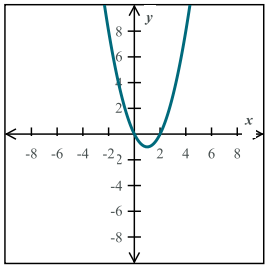
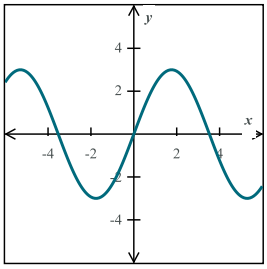


range: _____

domain: _____

Question 23 of 39

For each function graphed below, state whether it is one-to-one.

	Graph 1	Graph 2	Graph 3
			
One-to-one?	○ Yes ○ No	○ Yes ○ No	○ Yes ○ No
	Graph 4	**Graph 5**	**Graph 6**
			
One-to-one?	○ Yes ○ No	○ Yes ○ No	○ Yes ○ No

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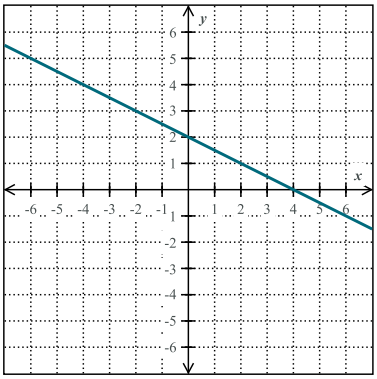
Question 24 of 39

Find the x-intercept and the y-intercept of the line below.



Question 25 of 39

For each equation, determine whether it is linear.

Equation	Is the equation linear?	
	Yes	No
(a) $y=4x$		
(b) $y=x-7$		
(c) $y=-x^3$		
(d) $y=2x^2-1$		

Question 26 of 39

For each equation, determine whether it is linear.

Equation	Is the equation linear?	
	Yes	No
(a) $y=4xy-7$		
(b) $x-\frac{5}{y}=0$		
(c) $9x+6+8y=x-5$		
(d) $y^2-8x=3$		

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Question 27 of 39

For each function, state whether it is linear.

Function 1	Function 2
$\{(3, 0), (6, 5), (9, 10), (12, 15)\}$ ○ Linear ○ Not linear	$\{(-4, -2), (-2, 1), (0, 0), (2, -1)\}$ ○ Linear ○ Not linear
Function 3	**Function 4**
x / y -3 / 2 -2 / 4 -1 / 8 0 / 16 ○ Linear ○ Not linear	x / y 5 / -4 9 / -9 13 / -15 17 / -23 ○ Linear ○ Not linear

Question 28 of 39

A table of values of a linear function is shown below.

x	-1	0	1	2
y	4	-1	-6	-11

Find the y-intercept and slope of the function's graph.

y-intercept: _____

slope: _____

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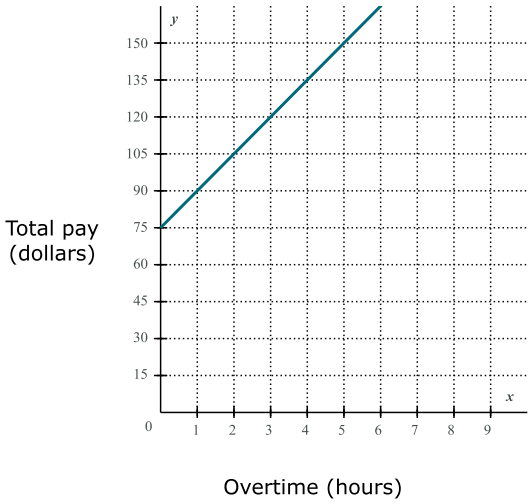
Question 29 of 39

Each day, Ivanna earns a fixed wage plus extra money for every hour of overtime she works. The graph shows her total pay (in dollars) versus the amount of overtime (in hours) that she works.



(a) Choose the statement that best describes how the amount of overtime and total pay are related. Then fill in the blank.

- ○ As the amount of overtime increases, the total pay decreases.

  At what rate is the total pay decreasing?

  $_____ per hour

- ○ As the amount of overtime increases, the total pay increases.

  At what rate is the total pay increasing?

  $_____ per hour

(b) What is Ivanna's total pay with 0 hours of overtime?

$_____

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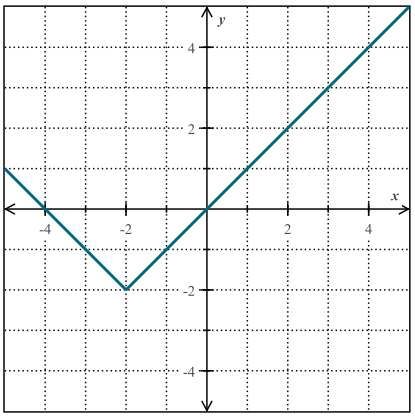
Question 30 of 39

The graph of a function is given below.
Give all y-intercepts and x-intercepts shown.



If there is more than one answer, separate them with commas.

Click on "None" if applicable.

y-intercept(s): _____

x-intercept(s): _____

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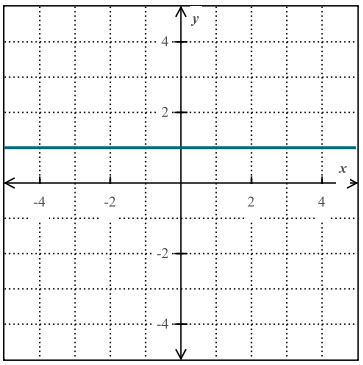
Question 31 of 39

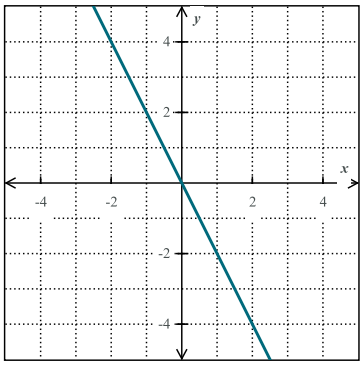
For each function graphed below, determine whether it is linear.

Function 1



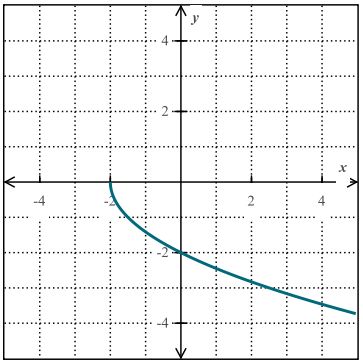
- Linear
- Not linear

Function 2



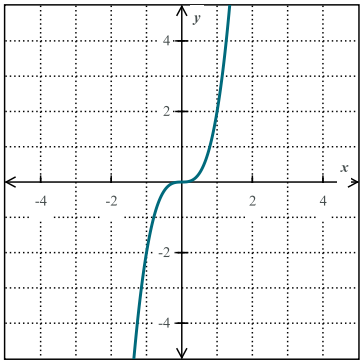
- Linear
- Not linear

Function 3



- Linear
- Not linear

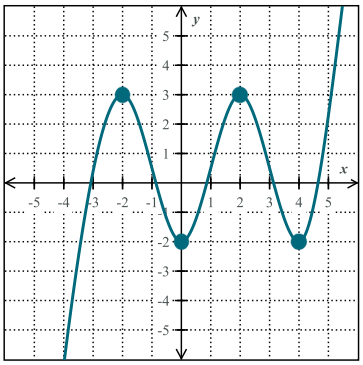
Function 4



- Linear
- Not linear

Question 32 of 39

Here is a graph of the function f.



Use the graph to find the following.
If there is more than one answer, separate them with commas.

All local minimum values of f: _____

All values at which f has a local minimum: _____

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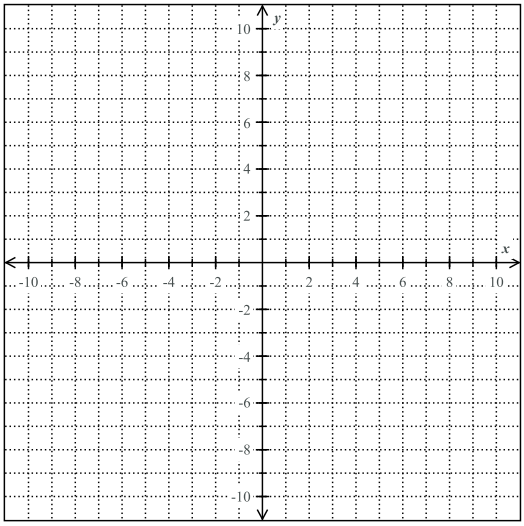
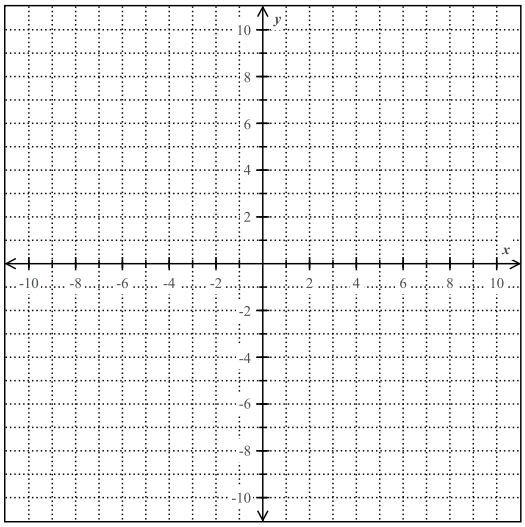
Question 33 of 39

Graph the line whose x-intercept is -1 and whose y-intercept is 1.





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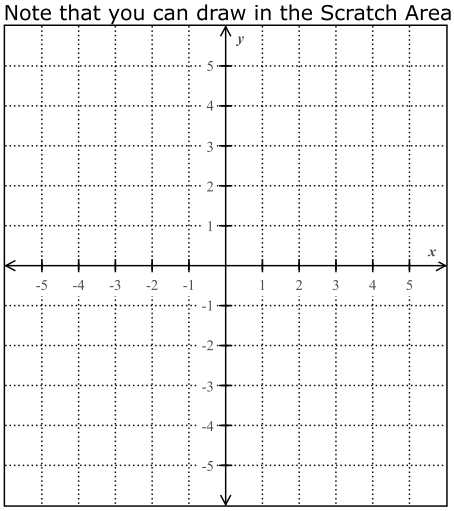
Question 34 of 39

Complete each statement below.
Note that you can draw in the Scratch Area, but it is not part of the answer.



(a) Complete the statements below about the graphs of $y = x - 4$ and $y = x$.

Compared to the graph of $y = x$, the graph of $y = x - 4$ is _____
(Blank 1)

Compared to the graph of $y = x$, the graph of $y = x - 4$ intersects the y-axis at _____
(Blank 2)

(b) Complete the statements below about the graphs of $y = \frac{5}{2}x$ and $y = x$.

Compared to the graph of $y = x$, the graph of $y = \frac{5}{2}x$ is _____
(Blank 3)

Compared to the graph of $y = x$, the graph of $y = \frac{5}{2}x$ intersects the y-axis at _____
(Blank 4)

Blank 1 Options	Blank 2 Options	Blank 3 Options	Blank 4 Options
steeper.	a higher point.	steeper.	a higher point.
equally steep.	the same point.	equally steep.	the same point.
less steep.	a lower point.	less steep.	a lower point.

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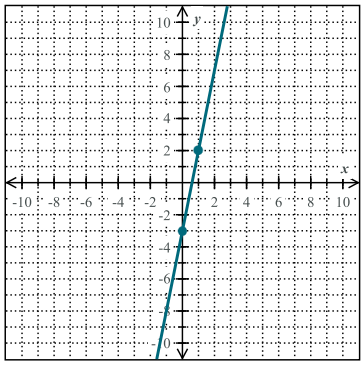
Question 35 of 39

Four different linear functions are represented below.

Function 1



Function 2

x	y
-2	3
-1	2
0	1
1	0
2	-1

Function 3

$y = -3x - 5$

Function 4

The slope is 2 and the y-intercept is 4.

Answer the following questions.

(a) Which functions have graphs with y-intercepts greater than 2? (Check all that apply.)

- ☐ Function 1
- ☐ Function 2
- ☐ Function 3
- ☐ Function 4

(b) Which function has the graph with a y-intercept closest to 0?

- ○ Function 1
- ○ Function 2
- ○ Function 3
- ○ Function 4

(c) Which function has the graph with the greatest slope?

- ○ Function 1
- ○ Function 2
- ○ Function 3
- ○ Function 4

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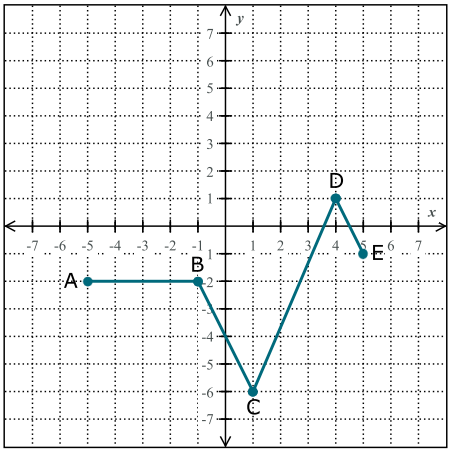
Question 36 of 39

Use the graph to determine the behavior of the function between the indicated points.



Points	Behavior of the function
A and B	○ increasing ○ constant ○ decreasing
B and C	○ increasing ○ constant ○ decreasing
C and D	○ increasing ○ constant ○ decreasing
D and E	○ increasing ○ constant ○ decreasing

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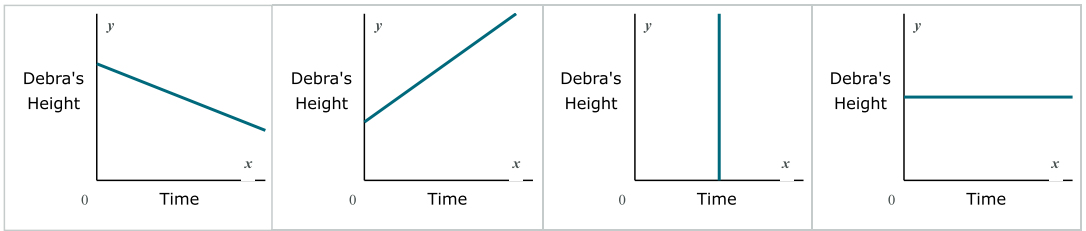
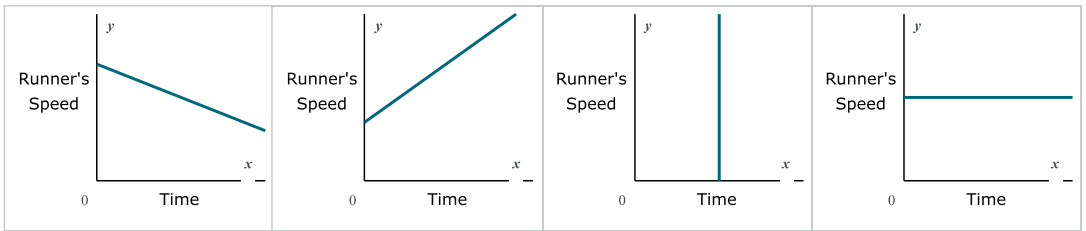
Question 37 of 39

For each scenario below, choose the best graph.

(a) Debra remains the same height for years.



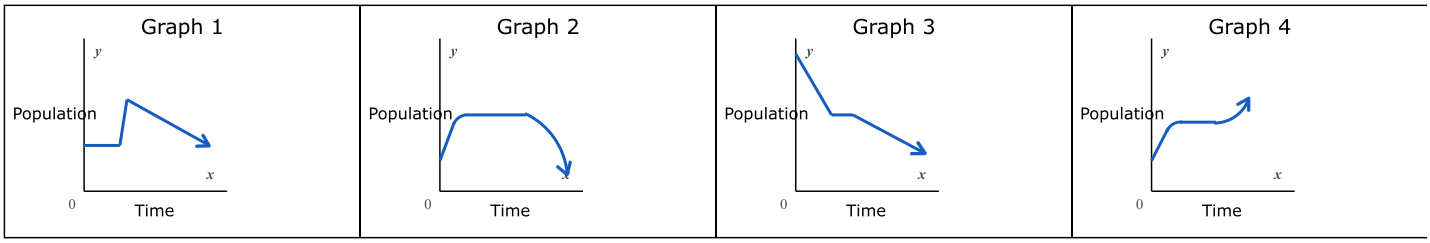
(b) At the end of a race, a runner slows down.



Question 38 of 39

(a) The population of a large city grows very quickly for several years. Then the growth begins to slow, and the population remains constant for several years. After this, the population grows again, but slower than before.

Navigate through the tabs below to see each table.

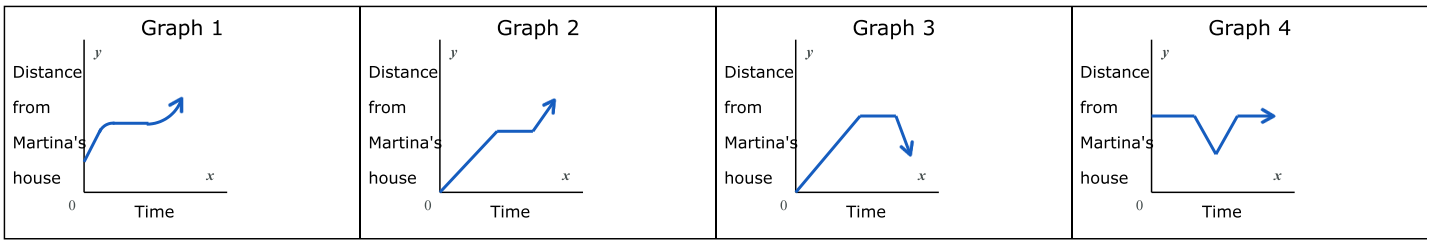


Choose the best graph.

Graph 1 Graph 2 Graph 3 Graph 4

(b) Martina leaves her house on her bike. She rides at a constant speed until she reaches a lemonade stand, where she parks her bike and takes a rest. Then she turns around and bikes home as fast as she can.

Navigate through the tabs below to see each table.



Choose the best graph.

Graph 1 Graph 2 Graph 3 Graph 4

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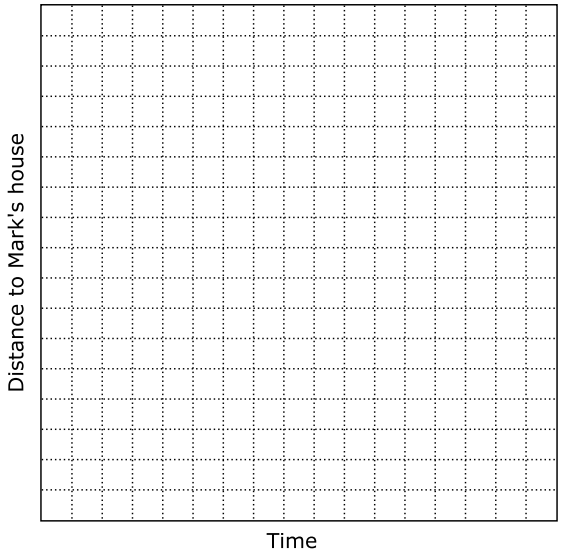
Question 39 of 39

Dante is delivering a pizza to Mark's house and drives at a constant speed toward the house. Then, he hits a traffic jam and stops for several minutes. After, he starts up again and drives toward the house.

Using three segments, sketch a possible graph that shows Dante's distance to Mark's house versus time.



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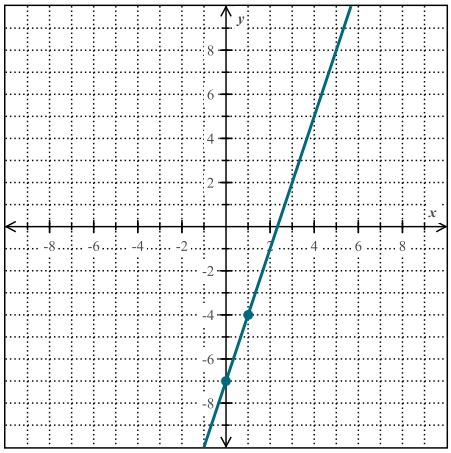
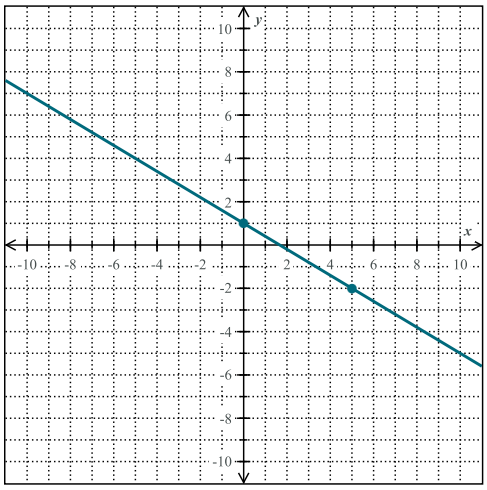
Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **39**

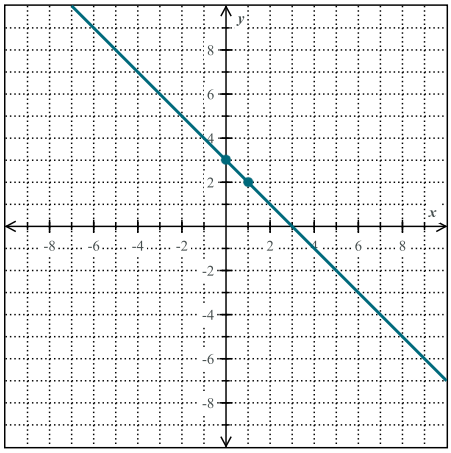
Question 1 of 39



Question 2 of 39



Question 3 of 39



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Question 4 of 39

(a) y-intercept: -4
(b) x-intercept: None

Question 5 of 39

x-intercept: -2

y-intercept: -5

Question 6 of 39

x-intercept(s): -1 , 3
y-intercept(s): 3

Question 7 of 39

$\{x \mid -5 \le x \le 3\} = [-5,\ 3]$

Question 8 of 39

- [ ] Graph 1
- [ ] Graph 2
- [x] Graph 3
- [ ] Graph 4
- [ ] Graph 5
- [x] Graph 6
- [ ] None of the graphs

Question 9 of 39

domain: $\{7, 9, -9\}$

range: $\{-2, 0\}$

Question 10 of 39

x	$f(x)$
-3	7
0	-5
3	-17
4	-21
5	-25

Question 11 of 39

$f(-5) = -11$
$g(-2) = 14$

Question 12 of 39

$h(5x) = 25x^2 + 7$

Question 13 of 39

79 miles

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Question 14 of 39

19.75 dollars

Question 15 of 39

(a) How much money does Goran still need after 7 weeks?

$\$567$

(b) If Goran still needs $\$339$, how many weeks has he been saving?

19 weeks

Question 16 of 39

	Description of Values	Set of Values
Domain:	◉ number of days the crew has worked ○ length of the road (in miles)	the set of all real numbers from 0 to 70
Range:	○ number of days the crew has worked ◉ length of the road (in miles)	the set of all real numbers from 300 to 510

Question 17 of 39

$h(-1)=0$

Question 18 of 39

domain $=\{2,\ -3,\ 0\}$
range $=\{1,\ 4,\ -3,\ -1\}$

Question 19 of 39

(a) Sam is 800 meter(s) from the lodge at the start of the ride. He is 0 meter(s) from the lodge at the end of the ride.

Each value in the [range] is a distance from the lodge.

(b) Sam starts the ride when the amount of time is 0 second(s). He continues riding until the amount of time is 160 second(s).

Each value in the [domain] is an amount of time since the ride started.

Question 20 of 39

Write the domain and range of f using interval notation.

domain: $(-4, 4)$

range: $(-5, 4)$

Question 21 of 39

domain $=(-4,\ -2)\cup(0,\ 1)$
range $=(-5,\ 4)$

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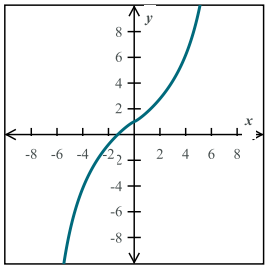
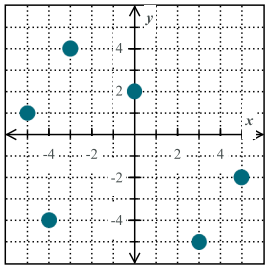
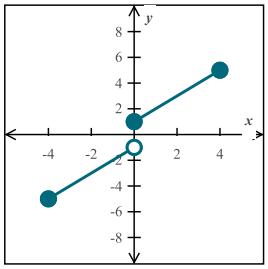
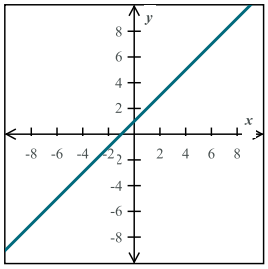
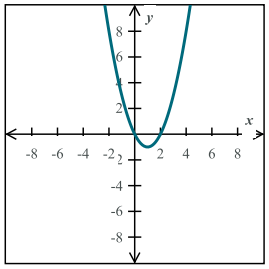
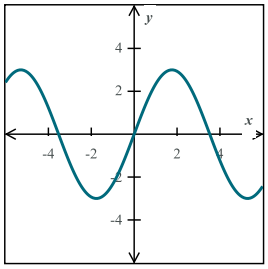
Question 22 of 39

(a) range: $y \leq -2$

(b) domain: All reals

Question 23 of 39

	Graph 1	Graph 2	Graph 3
			
One-to-one?	◉ Yes ○ No	◉ Yes ○ No	◉ Yes ○ No
	Graph 4	**Graph 5**	**Graph 6**
			
One-to-one?	◉ Yes ○ No	○ Yes ◉ No	○ Yes ◉ No

Question 24 of 39

(a) x-intercept: 4

(b) y-intercept: 2

Question 25 of 39

Equation	Is the equation linear?	
	Yes	No
(a) $y = 4x$	◉	○
(b) $y = x - 7$	◉	○
(c) $y = -x^3$	○	◉
(d) $y = 2x^2 - 1$	○	◉

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Question 26 of 39

Equation	Is the equation linear?	
	Yes	No
(a) $y=4xy-7$	○	◉
(b) $x-\frac{5}{y}=0$	○	◉
(c) $9x+6+8y=x-5$	◉	○
(d) $y^2-8x=3$	○	◉

Question 27 of 39

Function 1

$\{(3,\ 0),\ (6,\ 5),\ (9,\ 10),\ (12,\ 15)\}$

◉ Linear
○ Not linear

Function 2

$\{(-4,\ -2),\ (-2,\ 1),\ (0,\ 0),\ (2,\ -1)\}$

○ Linear
◉ Not linear

Function 3

x	y
-3	2
-2	4
-1	8
0	16

○ Linear
◉ Not linear

Function 4

x	y
5	-4
9	-9
13	-15
17	-23

○ Linear
◉ Not linear

Question 28 of 39

y-intercept: -1

slope: -5

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Question 29 of 39

(a) Choose the statement that best describes how the amount of overtime and total pay are related. Then fill in the blank.

- ○ As the amount of overtime increases, the total pay decreases.

  At what rate is the total pay decreasing?

  $_____ per hour

- ◉ As the amount of overtime increases, the total pay increases.

  At what rate is the total pay increasing?

  $15 per hour

(b) What is Ivanna's total pay with 0 hours of overtime?

$75

Question 30 of 39

y-intercept(s): 0

x-intercept(s): -4 , 0

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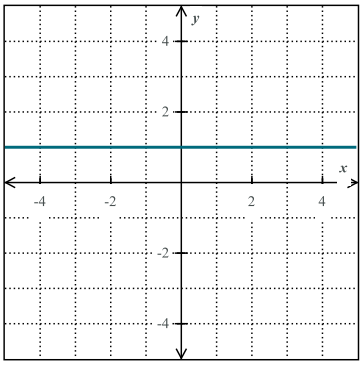
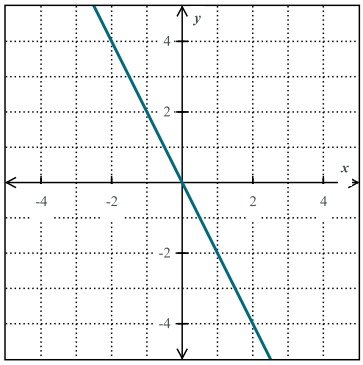
Question 31 of 39

Function 1



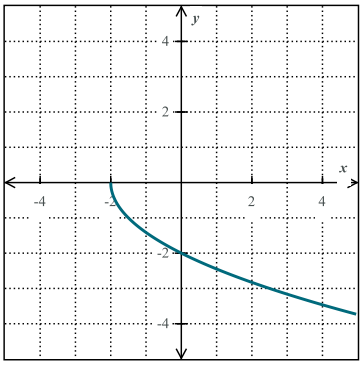
- (●) Linear
- ( ) Not linear

Function 2



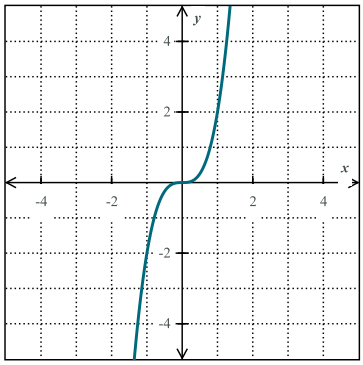
- (●) Linear
- ( ) Not linear

Function 3



- ( ) Linear
- (●) Not linear

Function 4



- ( ) Linear
- (●) Not linear

Question 32 of 39

All local minimum values of f: -2

All values at which f has a local minimum: $0, 4$

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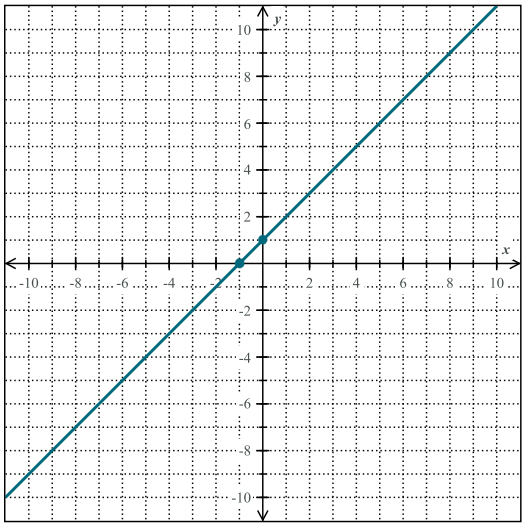
Question 33 of 39



Question 34 of 39

(a) Complete the statements below about the graphs of $y=x-4$ and $y=x$.

Compared to the graph of $y=x$, the graph of $y=x-4$ is equally steep.

Compared to the graph of $y=x$, the graph of $y=x-4$ intersects the y-axis at a lower point.

(b) Complete the statements below about the graphs of $y=\frac{5}{2}x$ and $y=x$.

Compared to the graph of $y=x$, the graph of $y=\frac{5}{2}x$ is steeper.

Compared to the graph of $y=x$, the graph of $y=\frac{5}{2}x$ intersects the y-axis at the same point.

Question 35 of 39

(a) Which functions have graphs with y-intercepts greater than 2?
(Check all that apply.)

- [ ] Function 1
- [ ] Function 2
- [ ] Function 3
- [x] Function 4

(b) Which function has the graph with a y-intercept closest to 0?

- ○ Function 1
- ◉ Function 2
- ○ Function 3
- ○ Function 4

(c) Which function has the graph with the greatest slope?

- ◉ Function 1
- ○ Function 2
- ○ Function 3
- ○ Function 4

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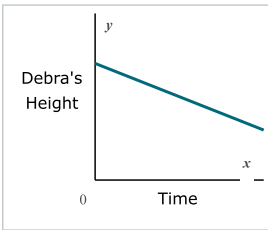
Question 36 of 39

Points	Behavior of the function
A and B	○ increasing ◉ constant ○ decreasing
B and C	○ increasing ○ constant ◉ decreasing
C and D	◉ increasing ○ constant ○ decreasing
D and E	○ increasing ○ constant ◉ decreasing

Question 37 of 39

(a) Debra remains the same height for years.



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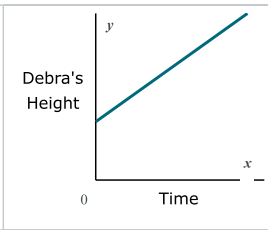

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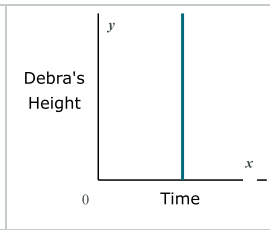

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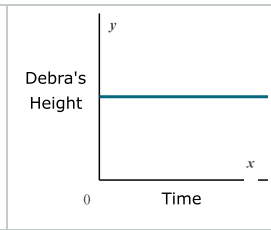
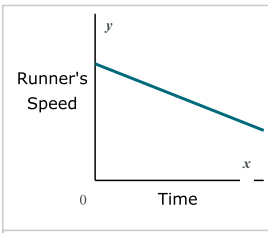


◉

(b) At the end of a race, a runner slows down.



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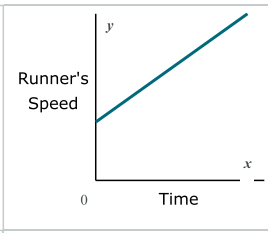

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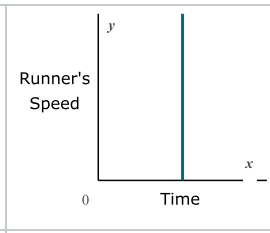

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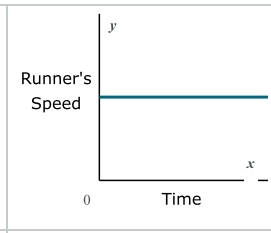

○

Question 38 of 39

(a) ○ Graph 1 ○ Graph 2 ○ Graph 3 ◉ Graph 4

(b) ○ Graph 1 ○ Graph 2 ◉ Graph 3 ○ Graph 4

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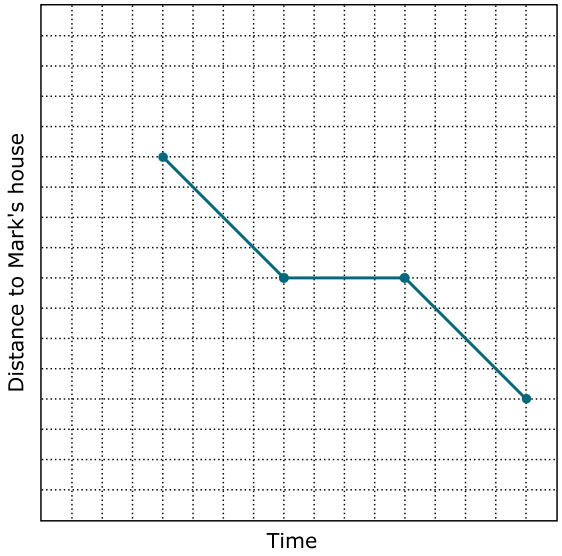
Question 39 of 39



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